

Quick Activity 2: Canteen Group Order

A canteen sells **rice meals at ₱85 each** and **drinks at ₱25 each**. Every order receives a **10% discount on the rice-meal subtotal only**. The canteen also charges a **₱5 packaging fee per rice meal**.

Develop an algorithm that calculates the final bill and the customer's change.

INPUT	PROCESS	OUTPUT
Quantity of rice meals, quantity of drinks, packaging fee, discount on rice meals, payment amount	Subtotal = quantity of rice meals + packaging fee Total of rice meals = subtotal of quantity of rice meals × discount Total cost = total cost of rice meals + quantity of drinks Change = payment - total cost	Total cost, change

DECOMPOSITION:

- Get the quantity of rice meals
- Get the quantity of drinks
- Get the packaging fee
- Get the discount
- Calculate the quantity of rice meals
- Calculate the packaging fee to the subtotal of rice meals
- Calculate the discount to the subtotal
- Get the overall cost of rice meals
- Calculate the quantity of drinks
- Calculate the total cost
- Calculate the change
- Get the total cost
- Get the change

PSEUDOCODE:

START

```
READ rice meals, drinks, payment amount
ricemealSubtotal ← ricemeals × 85
discount ← ricemealSubtotal × 0.10
discountedSubtotal ← ricemealSubtotal - discount
penSubtotal ← pens × 12
packagingfee ← ricemeals × 5
totalCost ← discountedSubtotal + drinkSubtotal + packagingfee
Change ← payment - totalCost
DISPLAY totalCost, change
```

END

DRY-RUN CALCULATION: QUICK ACTIVITY TWO

CASE ONE TEST:

A canteen sells rice meals at **₱85** each and drinks at **₱25** each. Every order receives a **10%** discount on the rice-meal subtotal only. The canteen also charges a **₱5** packaging fee per rice meal. The payment is **₱500**.

STEP	VARIABLE	CALCULATION	RESULT
1	ricemealSubtotal	3×85	₱255.00
2	drinkSubtotal	4×25	₱100.00
3	discount	255×0.10	₱25.50
4	packagingfee	25.50×5	₱127.50
5	totalCost	$127.50 + 100.00$	₱227.50
6	change	$500 - 227.50$	₱272.50

EXPECTED OUTPUT:

Total Cost = ₱227.50

Change = ₱272.50

CASE TWO TEST:

A canteen sells rice meals at **₱85** each and drinks at **₱25** each. Every order receives a **10%** discount on the rice-meal subtotal only. The canteen also charges a **₱5** packaging fee per rice meal. The payment is **₱600**.

STEP	VARIABLE	CALCULATION	RESULT
1	ricemealSubtotal	5×85	₱425.00
2	drinkSubtotal	3×25	₱75.00
3	discount	425×0.10	₱42.50
4	packagingfee	42.50×5	₱212.50
5	totalCost	$212.50 + 75.00$	₱287.50
6	change	$600 - 287.50$	₱312.50

EXPECTED OUTPUT:

Total Cost = ₱287.50

Change = ₱312.50